Exercise 1: Control Structures

Tables:

Customers

	CUSTOMERID	NAME	DOB	BALANCE	LASTMODIFIED	ISVIP
1	1	John Doe	5/15/1955, 12:00:0(	15000	6/23/2026, 12:17:2(	YES
2	2	Jane Smith	7/20/1990, 12:00:0(	5000	6/23/2026, 12:17:2(	*(null)*

Accounts

	ACCOUNTID	CUSTOMERID	ACCOUNTTYPE	BALANCE	LASTMODIFIED
1	1	1	Savings	1000	6/23/2026, 12:17:2:
2	2	2	Checking	1500	6/23/2026, 12:17:2:

Loans

	LOANID	CUSTOMERID	LOANAMOUNT	INTERESTRATE	STARTDATE	ENDDATE
1	1	1	5000	4	6/23/2026, 12:17:2:	7/13/2026, 12:17:2:
2	2	2	7000	6	6/23/2026, 12:17:2:	7/18/2026, 12:17:2:

Scenario 1: The bank wants to apply a discount to loan interest rates for customers above 60 years old.

Question: Write a PL/SQL block that loops through all customers, checks their age, and if they are above 60, apply a 1% discount to their current loan interest rates.

```sql
DECLARE
    age_value NUMBER;
BEGIN
    FOR person IN (
        SELECT CustomerID,DOB
        FROM Customers
    )
    LOOP
        age_value := TRUNC((SYSDATE - person.DOB)/365);

        IF age_value > 60 THEN
            UPDATE Loans
            SET InterestRate = InterestRate - 1
            WHERE CustomerID = person.CustomerID;
        END IF;
    END LOOP;

    COMMIT;
END;
/
```

Output:

```sql
SELECT CustomerID,
       InterestRate
FROM Loans;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download ▾ Execution time: 0.005 seconds

	CUSTOMERID	INTERESTRATE
1	1	4
2	2	6

Scenario 2: Scenario 2: A customer can be promoted to VIP status based on their balance.

Question: Write a PL/SQL block that iterates through all customers and sets a flag IsVIP to TRUE for those with a balance over $10,000.

```sql
SET SERVEROUTPUT ON;

BEGIN

    FOR data IN (
        SELECT CustomerID,Balance
        FROM Customers
    )
    LOOP

        IF data.Balance > 10000 THEN

            UPDATE Customers
            SET IsVIP = 'YES'
            WHERE CustomerID = data.CustomerID;

            DBMS_OUTPUT.PUT_LINE(
                'VIP status granted to '
                || data.CustomerID
            );

        END IF;

    END LOOP;

    COMMIT;

END;
/
```

Output:

```sql
SELECT CustomerID,
       Name,
       IsVIP
FROM Customers;
```

Query result   Script output   DBMS output   Explain Plan

Download   Execution time: 0.006 seconds

	CUSTOMERID	NAME	ISVIP
1	1	John Doe	YES
2	2	Jane Smith	*(null)*

__**Scenario 3: The bank wants to send reminders to customers whose loans are due within the next 30 days.**__

Question: Write a PL/SQL block that fetches all loans due in the next 30 days and prints a reminder message for each customer.

```sql
SET SERVEROUTPUT ON;

BEGIN

    FOR loan_data IN (
        SELECT c.Name,
               l.LoanID
        FROM Customers c,
             Loans l
        WHERE c.CustomerID = l.CustomerID
        AND l.EndDate <= SYSDATE + 30
    )
    LOOP

        DBMS_OUTPUT.PUT_LINE(
            'Reminder sent to '
            || loan_data.Name
        );

    END LOOP;

END;
/
```

__**Output:**__

```sql
SELECT LoanID,
       CustomerID,
       EndDate
FROM Loans
WHERE EndDate <= SYSDATE + 30;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.006 seconds

	LOANID	CUSTOMERID	ENDDATE
1	1	1	7/13/2026, 12:17:2:
2	2	2	7/18/2026, 12:17:2: